

ZHONGNING, China

WMO: 537050

Lat: 37.4843N

Lon: 105.6837E

Elev: 3914

StdP: 12.73

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.6	(c) 8.3	(d) -13.2	(e) 3.1	(f) 15.6	(g) -9.8	(h) 3.7	(i) 18.5	(j) 20.1	(k) 30.8	(l) 18.1	(m) 26.9	(n) 4.5	(o) 220	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 92.8	(d) 65.9	(e) 90.4	(f) 65.2	(g) 87.9	(h) 64.3	(i) 71.2	(j) 85.1	(k) 69.7	(l) 83.2	(m) 68.3	(n) 81.3	(o) 6.4	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	116.5	77.5	65.6	109.4	76.6	64.0	103.2	75.5	38.0	85.3	36.5	83.2	35.2	81.1	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.1	(b) 16.9	(c) 14.4	(e) -0.3	(f) 97.5	(g) 4.4	(h) 2.2	(i) -3.5	(j) 99.1	(k) -6.1	(l) 100.4	(m) -8.5	(n) 101.6	(o) -11.7	(p) 103.2		
(a) 20.1	(b) 16.9	(c) 14.4	(e) -1.7	(f) 73.6	(g) 4.1	(h) 2.0	(i) -4.6	(j) 75.0	(k) -7.0	(l) 76.2	(m) -9.3	(n) 77.3	(o) -12.2	(p) 78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	22.8	30.7	43.9	56.5	65.4	73.2	76.7	73.7	64.6	52.5	38.8	26.4		
	DBStd	19.69	6.72	8.21	8.61	8.02	6.65	5.37	5.45	5.67	6.10	6.89	6.93	6.29		
	HDD50	2763	842	540	226	29	1	0	0	0	1	53	339	732		
	HDD65	5736	1307	959	656	272	77	6	1	5	79	390	787	1197		
	CDD50	3569	0	1	35	225	478	696	829	735	438	130	2	0		
	CDD65	1067	0	0	0	18	90	252	364	275	66	2	0	0		
	CDH74	10367	0	0	15	227	975	2481	3660	2443	531	35	0	0		
	CDH80	3887	0	0	1	54	270	942	1585	915	119	1	0	0		
(14) Wind	WSAvg	6.3	6.2	6.5	7.3	7.5	7.0	6.3	6.0	5.7	5.5	5.3	5.7	6.0		
(15) Precipitation	PrecAvg	8.5	0.0	0.1	0.2	0.5	0.9	0.9	1.6	2.3	1.2	0.6	0.2	0.0		
	PrecMax	15.4	0.3	0.4	0.7	1.8	2.7	2.5	4.6	6.3	3.4	1.7	0.9	0.1		
	PrecMin	3.3	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.5	0.1	0.0	0.0	0.0		
	PrecStd	3.1	0.1	0.1	0.2	0.5	0.7	0.6	1.1	1.7	0.8	0.4	0.2	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	60.7	75.5	86.4	89.3	93.9	97.2	94.6	88.5	78.9	64.8	49.9		
		MCWB	33.9	41.5	49.2	55.8	58.2	62.3	67.8	68.4	63.6	56.7	46.9	35.9		
	2%	DB	42.8	53.2	67.8	80.1	86.2	91.1	93.6	91.1	83.3	73.8	59.9	44.9		
		MCWB	30.6	37.4	45.7	53.0	57.7	62.5	67.3	67.2	61.3	54.3	44.2	33.0		
	5%	DB	38.5	48.9	63.2	75.7	83.1	88.5	91.0	88.2	80.0	69.7	55.8	41.1		
		MCWB	28.2	35.4	43.8	51.0	56.6	62.1	66.9	66.4	60.4	52.7	42.2	30.9		
	10%	DB	34.6	45.0	58.8	71.6	79.7	85.7	88.3	85.4	76.6	65.8	51.4	37.6		
		MCWB	25.8	33.0	41.6	49.5	55.6	61.6	66.6	65.7	59.7	50.5	40.1	28.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	42.0	50.4	58.6	63.4	69.1	74.4	73.2	68.3	58.8	48.4	36.9		
		MCDB	47.5	57.7	70.9	77.4	79.3	84.1	88.5	87.3	78.3	73.1	61.0	47.1		
	2%	WB	31.0	38.4	47.1	55.5	60.9	67.1	71.9	71.1	65.3	56.5	45.6	33.3		
		MCDB	42.4	51.5	64.8	73.6	77.5	82.5	86.5	84.3	76.4	69.6	57.0	43.4		
	5%	WB	28.4	35.9	44.7	53.2	59.2	65.6	70.2	69.7	63.4	54.7	43.2	31.1		
		MCDB	38.1	47.8	60.8	71.2	77.3	80.6	84.4	82.4	74.0	66.2	53.4	40.3		
	10%	WB	25.9	33.4	42.4	51.2	57.6	64.2	68.8	68.2	61.7	52.5	40.7	28.9		
		MCDB	34.0	44.5	57.7	67.9	74.8	79.2	82.5	80.4	72.3	63.1	50.5	37.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	22.4	22.5	23.0	22.6	21.2	19.3	18.9	19.5	21.8	20.0	19.1		
		MCDBR	27.0	28.6	29.0	28.7	28.2	25.7	23.1	23.1	25.1	26.5	26.8	23.9		
		MCWBR	17.6	16.8	14.1	11.2	10.1	8.2	6.9	7.4	8.5	10.8	14.2	15.2		
		MCDBR	26.6	27.5	26.7	23.6	22.6	20.2	18.6	18.8	18.5	23.2	24.8	23.0		
	5% WB	MCWBR	17.6	16.3	13.5	10.2	9.4	7.7	7.4	7.6	8.0	10.5	14.0	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.430	0.490	0.531	0.534	0.527	0.507	0.522	0.498	0.459	0.395	0.357		
	taud		2.005	1.870	1.768	1.714	1.772	1.850	1.977	1.914	1.912	1.924	1.998	2.060		
	Ebn at Noon		245	244	242	241	243	245	249	241	238	232	233	240		
	Edh at Noon		41	54	66	73	70	65	57	59	56	50	41	36		
(44) All-Sky Solar Radiation	RadAvg		887	1175	1560	1862	2056	2069	1959	1775	1426	1258	991	843		
	RadStd		93	84	82	100	100	122	108	111	109	98	76	59		

Historical Trends

DBAvg	Heating		Cooling			Degree-Days				
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	+0.97	N/A	+1.98	+1.22	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.76	N/A	N/A	+0.81	N/A	-1.24	N/A	N/A	+133	+72

Nomenclature: See separate page